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Experiment: 8

Question 1: Change Text Content

Problem:

Select a paragraph element and change its text content when a button is clicked.

Question 2: Toggle CSS Class

Problem:

Write a function to toggle a class ("active") on a div when it is clicked.

Question 3: Add List Item

Problem:

Add a new element to an unordered list when a button is clicked.

Question 4: Remove Element

Problem:

Remove a specific element from the DOM when a delete button is clicked.

Question 5: Input Value Display

Problem:

Display the value of an input field in a <p> tag in real-time as the user types.

Question 6: Change Image Source

Problem:

Change the src of an image when a button is clicked.

Question 7: Form Validation

Problem:

Validate a form to ensure the input field is not empty before submission. Show an error message if invalid.

Question 8: Event Delegation

Problem:

Add click events to multiple list items using event delegation (attach event to parent).

Question 9: Create Elements Dynamically

Problem:

Create a card (div with title + description) dynamically using JavaScript and append it to the DOM.

Question 10: Highlight Selected Item

Problem:

When a list item is clicked, highlight it and remove highlight from others.

SOLUTION:

CODE: [Index.html]

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>DOM Experiment Pro</title>

<link
href="https://fonts.googleapis.com/css2?family=Orbitron:wght@500&family=Poppins:wght@300;400&
display=swap" rel="stylesheet">

<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/fontawesome/6.5.0/css/all.min.css">

<link rel="stylesheet" href="style.css">
</head>
```

```

<body>
<div class="card">
  <h1><i class="fa-solid fa-code"></i> DOM Lab</h1>

  <select id="question" onchange="loadUI()">
    <option value="1">📝 Change Text</option>
    <option value="2">📝 Toggle Class</option>
    <option value="3">➕ Add List Item</option>
    <option value="4">✖ Remove Element</option>
    <option value="5">📝 Live Input</option>
    <option value="6">📝 Change Image</option>
    <option value="7">✅ Form Validation</option>
    <option value="8">📝 Event Delegation</option>
    <option value="9">📝 Create Card</option>
    <option value="10">📝 Highlight Item</option>
  </select>

  <p id="desc" class="desc"></p>

  <div id="dynamicArea" class="fade"></div>

  <div id="result"></div>
</div>

<script src="script.js"></script>
</body>
</html>

```

[Stye.css]:

```

body {
margin: 0;
height: 100vh;

```

```
display: flex;
justify-content:
center; align-
items: center;
background:
linear-
gradient(135deg
, #b7ccd1,
#dbe9ec);
font-family:
'Poppins', sans-
serif;
}
```

```
.card { background: #ececec; padding:
40px; border-radius: 20px; width: 420px;
text-align: center; box-shadow: 0 15px
40px rgba(0,0,0,0.2); animation: fadeIn
0.6s ease;
}
```

```
h1 { font-family: 'Orbitron', sans-
serif; color: #118c8f;
}
```

```
select, input, button {
width: 100%; padding:
12px; margin: 10px 0;
border-radius: 12px;
border: 2px solid #ff5a5a;
font-size: 14px; transition:
0.3s;
}
```

```
select:hover, input:hover {
transform: scale(1.02);
}
```

```
button { background: linear-gradient(90deg, #ff7a6b,
#ff5a5a); color: white; border: none; cursor:
pointer; transition: 0.3s;
}
```

```
button:hover {  
  transform: scale(1.05);  
}
```

```
button:active {  
  transform: scale(0.95);  
}
```

```
.desc {  font-size: 13px;  
  color: #555;  margin-  
  bottom: 10px;  
}
```

```
.fade {  animation: fadeIn  
0.5s ease;  
}
```

```
.active {  
  background: yellow;  
  transition: 0.3s;  
}
```

```
.highlight {  background:  
lightgreen;  
}
```

```
.card-box {  background:  
white;  padding: 10px;  
margin-top: 10px;  border-  
radius: 10px;  animation:  
fadeIn 0.4s ease;  
}
```

```
@keyframes fadeIn {  
  from { opacity: 0;  
  transform: translateY(10px); }  
  to { opacity: 1;  
  transform: translateY(0); }  
}
```

[Script.js]:

```
const descriptions = {  
  1: "Change text of a paragraph dynamically.",
```

```

2: "Toggle a CSS class on click.",
3: "Add your own list items.",
4: "Add and remove elements dynamically.",
5: "Display input value live.",
6: "Change image using URL.",
7: "Form validation (no empty input).",
8: "Click list items (event delegation).",
9: "Create custom cards.",
10: "Highlight selected item."
};

function loadUI() {  let q =
document.getElementById("question").value;  let area
= document.getElementById("dynamicArea");

    document.getElementById("desc").innerText = descriptions[q];

    switch(q) {

        // Q.1 Change Text
    case "1":
        area.innerHTML = `
            <input id="newText" placeholder="Enter new text">
            <p id="text">Original Text</p>
            <button onclick="changeText()">Change</button>`;
        break;

        // Q.2 Toggle Class
    case "2":
        area.innerHTML = `
            <div id="box" style="padding:20px;background:#ddd;cursor:pointer;">
                Click Me
            </div>
            <p id="msg"></p>
        `;

        document.getElementById("box").onclick = () => {
let box = document.getElementById("box");          let
msg = document.getElementById("msg");
box.classList.toggle("active");

```

```

        if (box.classList.contains("active")) {
box.style.background = "lightgreen";          msg.innerHTML
= "Toggled Successfully!";          msg.style.color =
"green";
        } else {
            box.style.background = "#ddd";
msg.innerHTML = "";
        }
    };
break;

```

```

// Q.3 Add List Item
case "3":
    area.innerHTML = `
        <input id="itemInput" placeholder="Enter item">
        <ul id="list"></ul>
        <button onclick="addItem()">Add Item</button>`;
break;

```

```

// Q.4 Remove Element (UPDATED)
case "4":
    area.innerHTML = `
        <input id="removeInput" placeholder="Enter item">
        <button onclick="addRemovableItem()">Add Item</button>
        <ul id="removeList"></ul>
    `;
break;

```

```

// Q.5 Live Input
case "5":
    area.innerHTML = `
        <input oninput="showValue(this.value)" placeholder="Type something">
        <p id="output"></p>`;
break;

```

```

// Q.6 Image Changer (UPDATED LIVE)
case "6":
    area.innerHTML = `
        <input id="imgUrl" placeholder="Enter image URL" oninput="liveImageChange()">
        
    `;
break;

```

```
// Q.7 Form Validation
```

```
case "7":
```

```
    area.innerHTML = `  
        <input id="name" placeholder="Enter name">  
        <button onclick="validate()">Submit</button>  
        <p id="error"></p>`;
```

```
break;
```

```
// Q.8 Event Delegation
```

```
case "8":
```

```
    area.innerHTML = `  
        <input id="itemVal" placeholder="Add item">  
        <button onclick="addDelegationItem()">Add</button>  
        <ul id="parent"></ul>  
        <p id="selected"></p>  
    `;
```

```
    document.getElementById("parent").onclick = e => {  
if(e.target.tagName === "LI") {  
document.getElementById("selected").innerText =  
    "You selected: " + e.target.innerText;  
    }  
};
```

```
break;
```

```
// Q.9 Create Card
```

```
case "9":
```

```
    area.innerHTML = `  
        <input id="title" placeholder="Enter title">  
        <input id="descInput" placeholder="Enter description">  
        <button onclick="createCard()">Create Card</button>  
        <div id="cards"></div>`;
```

```
break;
```

```
// Q.10 Highlight Item
```

```
case "10":
```

```
    area.innerHTML = `  
        <input id="highlightInput" placeholder="Add item">  
        <button onclick="addHighlightItem()">Add</button>        <ul id="highlightList"></ul>`;
```



```

        document.getElementById("highlightList").onclick = e => {
if(e.target.tagName === "LI") {
document.querySelectorAll("#highlightList li")
.forEach(li => li.classList.remove("highlight"));
e.target.classList.add("highlight");
        }
    };
break;
    }
}

```

```

function changeText() {
    let val =
document.getElementById("newText").value;
document.getElementById("text").innerText = val || "Empty!";
}

```

```

function addItem() {
    let val =
document.getElementById("itemInput").value;
    if (!val) return;

```

```

        let li = document.createElement("li");
        li.innerText
= val;
document.getElementById("list").appendChild(li);
document.getElementById("itemInput").value = "";
}

```

```

function addRemovableItem() {
    let val = document.getElementById("removeInput").value;
    if (!val) return;

```

```

        let li = document.createElement("li");
li.style.display = "flex";
li.style.justifyContent
= "space-between";
li.style.alignItems =
"center";

```

```

        let span = document.createElement("span");
span.innerText = val;

```

```

        let btn = document.createElement("button");
        btn.innerText = "Delete";
        btn.style.marginLeft = "10px";

```

```
btn.style.padding = "5px 10px";
btn.style.borderRadius = "8px";
btn.style.border = "none";
btn.style.background = "#ff5a5a";
btn.style.color = "white"; btn.style.cursor
= "pointer";
```

```
btn.onclick = () => li.remove();
```

```
li.appendChild(span);
li.appendChild(btn);
```

```
document.getElementById("removeList").appendChild(li);
document.getElementById("removeInput").value = "";
}
```

```
function showValue(val) {
document.getElementById("output").innerText = val;
}
```

```
function liveImageChange() { let url =
document.getElementById("imgUrl").value; let img
= document.getElementById("img");
```

```
if (!url) return;
```

```
img.src = url;
```

```
img.onerror = () => { img.src =
"https://via.placeholder.com/100";
};
}
```

```
function validate() { let val =
document.getElementById("name").value; let msg
= document.getElementById("error");
if (!val)
{
msg.innerText = "Field required!";
msg.style.color = "red";
} else {
```

```

    msg.innerText = "Form submitted successfully!";
    msg.style.color = "green";
  }
}

function addDelegationItem() { let val =
document.getElementById("itemVal").value; if (!val)
return;

  let li = document.createElement("li"); li.innerText =
val;
document.getElementById("parent").appendChild(li);
}

function createCard() { let title =
document.getElementById("title").value; let desc =
document.getElementById("descInput").value;

  if (!title || !desc) return;

  let div = document.createElement("div");
div.className = "card-box"; div.innerHTML =
`<h3>${title}</h3><p>${desc}</p>`;

  document.getElementById("cards").appendChild(div);
}

function addHighlightItem() {
  let val = document.getElementById("highlightInput").value;
if (!val) return;

  let li = document.createElement("li");
li.innerText = val;
  document.getElementById("highlightList").appendChild(li);
}

loadUI();

```

[OUTPUT]:

Sol 1: Change Text.

</> DOM Lab

📄 Change Text ▼

Change text of a paragraph dynamically.

Enter new text

Original Text

Change

</> DOM Lab

📄 Change Text ▼

Change text of a paragraph dynamically.

Divesh

Divesh

Change

Sol 2: Toggle Class.

</> DOM Lab

💡 Toggle Class ▼

Toggle a CSS class on click.

Click Me

</> DOM Lab

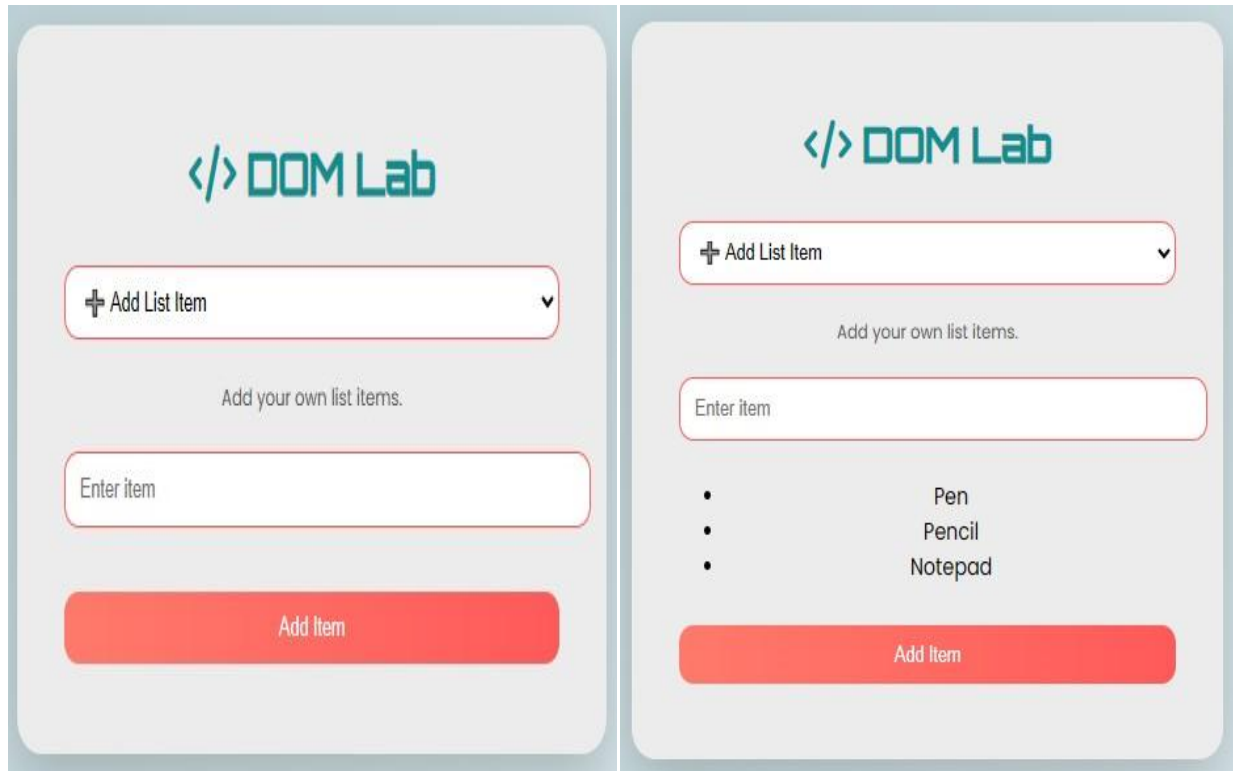
💡 Toggle Class ▼

Toggle a CSS class on click.

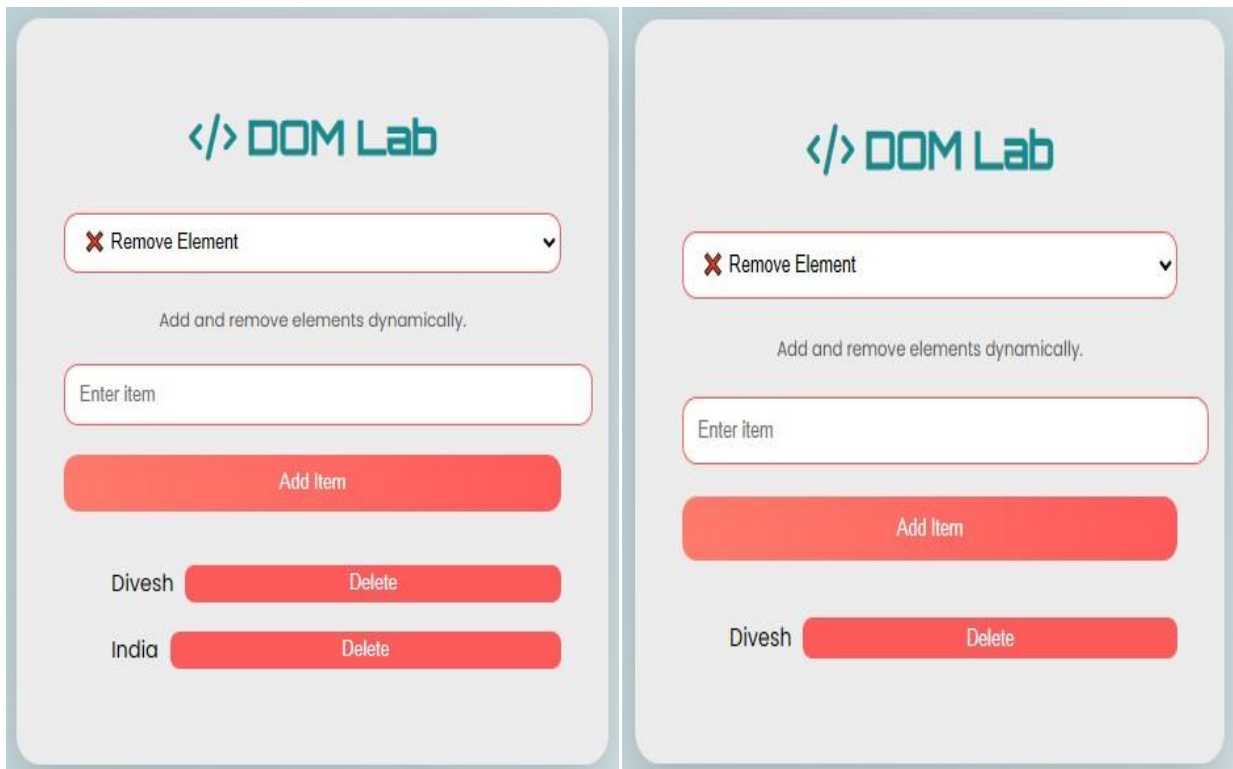
Click Me

Toggled Successfully!

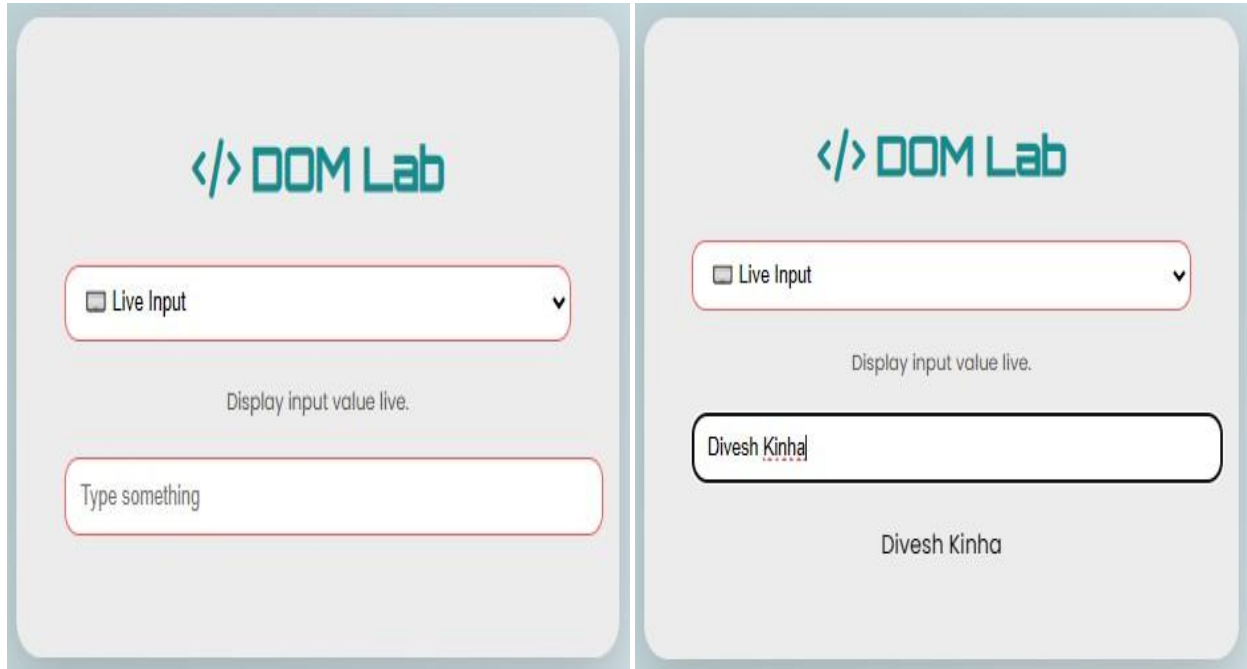
Sol 3: Add Item to List.



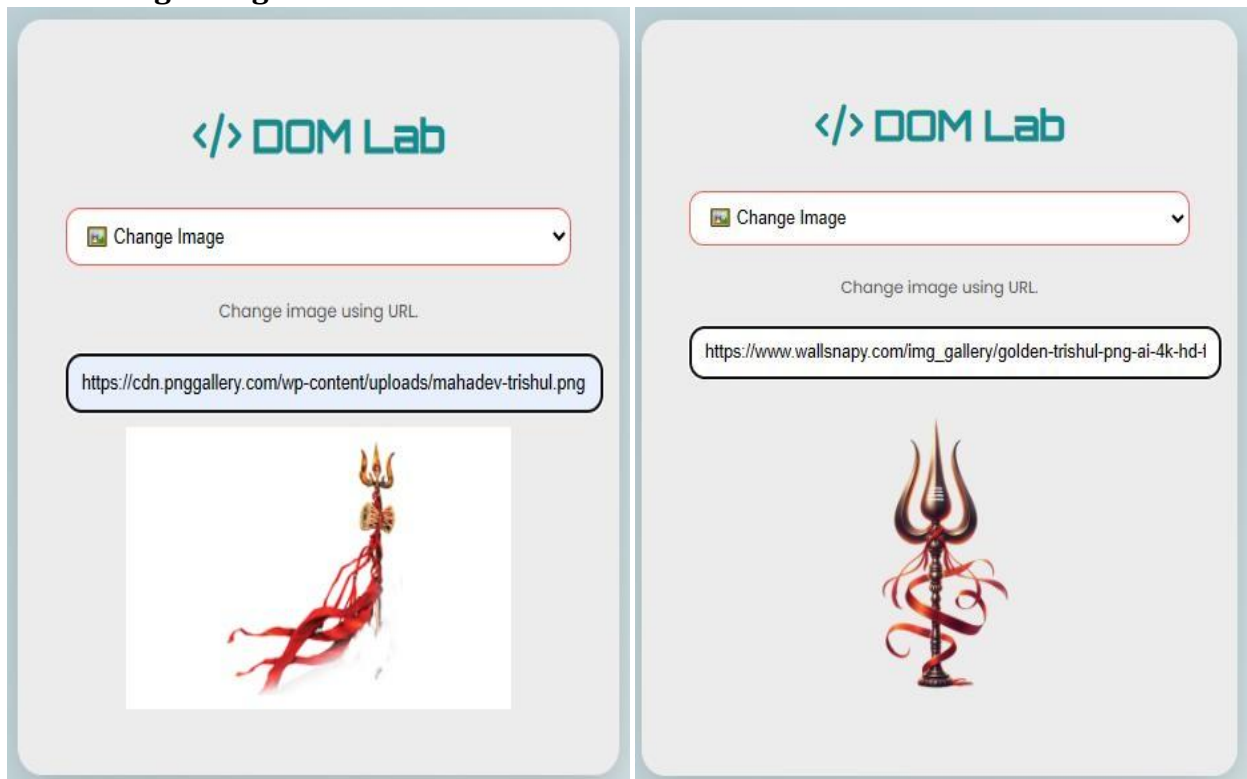
Sol 4: Remove Element.



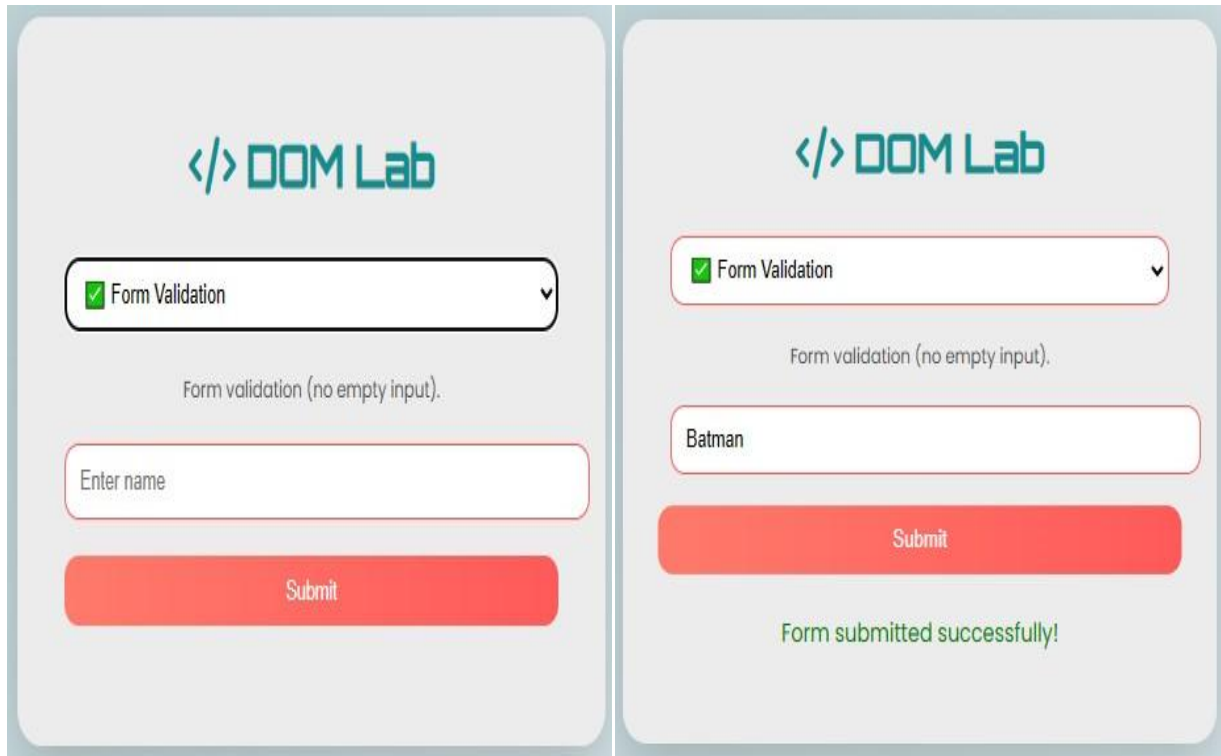
Sol 5: Live Input.



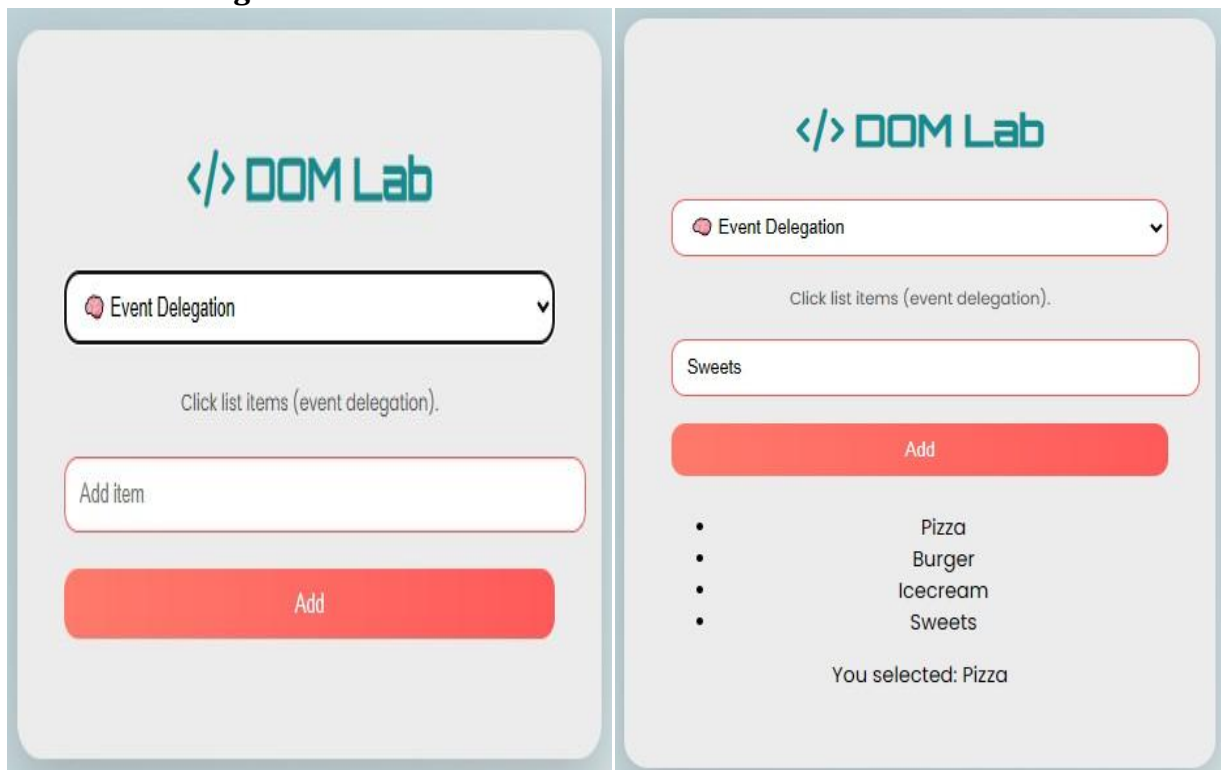
Sol 6: Change Image.



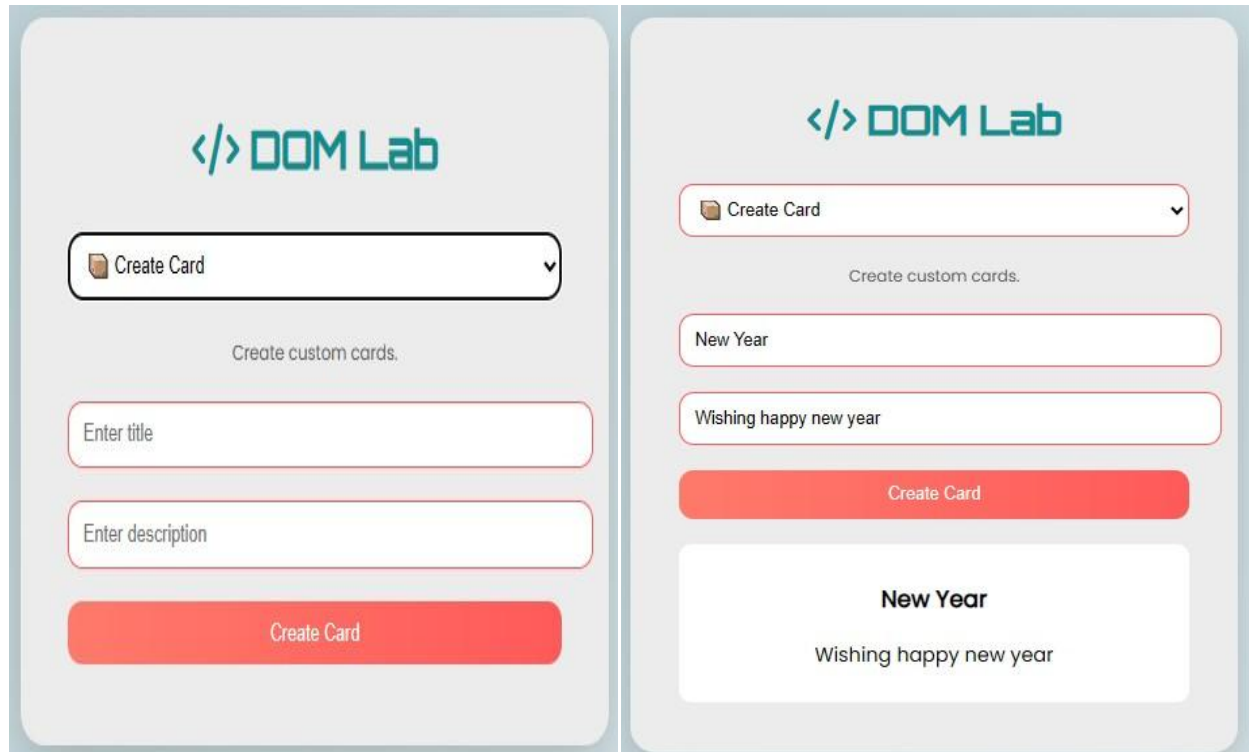
Sol 7: Form Validation.



Sol 8: Event Delegation.



Sol 9: Create Card.



Sol 10: Highlight Item.

